Brief A: Technical Communication

To: Lead Maintenance Engineer

From: Silas Kibet, Lead Data Scientist

Subject: Deployment of Predictive Maintenance (PdM) Vibration Classification Model

Overview We have developed a Logistic Regression classification model to monitor continuous sensor streams and predict mechanical degradation. The model processes raw time-series vibration data to identify anomalous behavior indicating an approaching failure state.

Sensor Logic & Feature Extraction The model does not ingest raw signal data directly. Instead, we apply a 15-step rolling window to extract three primary health features:

- **Rolling Mean:** To smooth high-frequency noise.
- **Max-Min Range:** To capture sudden amplitude spikes.
- **Root Mean Square (RMS):** To measure the overall energy of the vibration signal.

As mechanical wear increases, RMS exhibits an exponential upward trend, serving as our primary degradation indicator.

Model Performance & Metrics The model prioritizes sensitivity (Recall) over absolute precision. In our test environment, the model achieved a high recall rate for the "Failing" class. This threshold tuning was intentional: we aim to minimize False Negatives (missing a genuine failure, which results in a catastrophic breakdown). The trade-off is a slight increase in False Positives, which may trigger early maintenance alerts, but ensures component safety margins are never breached.

Brief B: Executive Communication

To: Chief Financial Officer (CFO)

From: Silas Kibet, Lead Data Scientist

Subject: Financial Impact and ROI of Predictive Maintenance Implementation

Overview Our team has successfully piloted a data-driven early warning system for our core machinery. By analyzing existing operational data, this system allows us to detect equipment wear and tear *before* a machine actually breaks down.

Value Proposition & Risk Reduction Currently, our maintenance strategy relies on either scheduled check-ups (which can be unnecessary and costly) or reactive repairs (which lead to expensive, unplanned downtime and halted production). This new model transitions us to a "predictive" strategy. We intervene only when the data indicates a repair is actually needed, drastically reducing the risk of sudden operational halts.

Cost Savings & ROI By catching anomalies early, we shift from purchasing expensive emergency replacement parts to utilizing standard preventative maintenance inventory. Based on initial testing, we estimate a significant reduction in unplanned downtime. The operational cost savings from avoiding just one catastrophic failure—including lost production hours and expedited shipping for replacement parts—more than covers the computational and development costs of this initiative, ensuring a rapid and compounding Return on Investment (ROI).

Reflection

Switching contexts between the two briefs highlighted the importance of translating data science outcomes into audience-specific value. Writing for the engineer required a focus on methodology, thresholds, and statistical trade-offs (like prioritizing recall over precision). Conversely, writing for the CFO required stripping away the mathematical mechanics entirely, focusing instead on how the model acts as a lever to reduce financial risk, optimize resource allocation, and protect the bottom line. It reinforced that a model's utility is only as good as our ability to communicate its business impact.